

Class XII | Unit 4 | weight 15%

File handling

Board weight about 15% | 5 theory periods and 5 lab periods

Variables disappear when the program ends. A file keeps data on secondary storage. This lecture opens files, chooses a mode, and reads and writes both text and binary data.

What you should be able to do

- Define file handling and distinguish text and binary files.
- Open a file for reading, writing or appending.
- Explain BOF and EOF, and single-character, string and binary streams.
- Write programs that create and read a text file and a binary file of student records.

Path through the lecture

- 1 Show a text file in an editor, then show that a binary file is not meant to be read that way.
- 2 Teach open, check, use, close as a fixed four-step habit.
- 3 Write the student text program, then the same student as a binary record.

4.1 Files, modes and ends

File handling. The part of a program that creates, writes, reads and updates data stored in a file on secondary storage.

A **text file** stores characters. You can open it in an editor and read the numbers and words. A **binary file** stores the bytes of the data as the program holds them in memory. It is compact and it preserves the exact bits of an int or a float. It is not a document for a person to edit.

Modes used with ifstream, ofstream and fstream

Mode	Effect
ios::in	Open for reading. The file should already exist.
ios::out	Open for writing. A new file is created. An existing file is emptied first.
ios::app	Open for appending. New output is added at the end. Existing contents stay.
ios::binary	Treat the file as binary. Combine it with in or out, for example ios::in ios::binary.

ofstream is for output, ifstream is for input, and fstream can do both. After open, test the stream before you use it. Close the file when the work is finished so the buffer is written and the file is released.

BOF. Beginning of file. The position before the first byte, where a newly opened input file starts.

EOF. End of file. The condition that becomes true when a read goes past the last data. The loop while (inFile >> word) stops when no more input can be read.

A stream is a flow of data. **Single-character I/O** uses get and put. **String I/O** uses >>, getline and <<. **Binary I/O** uses read and write with a memory address and a size. write(reinterpret_cast<char*>(&s), sizeof(s)) copies the bytes of the object s into the file. read copies them back into an object of the same type.

4.2 Text and binary programs

Program 4.1 Write and read a text file

```

#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {
    ofstream outFile("student.txt");
    outFile << "Ayesha" << endl;
    outFile << 88 << endl;
    outFile.close();

    ifstream inFile("student.txt");
    string name;
    int marks;
    inFile >> name >> marks;
    cout << "Read: " << name << endl;
    cout << "Read: " << marks << endl;
    inFile.close();
    return 0;
}

```

Program 4.1 creates student.txt, writes a name and a mark, closes it, then opens it again and reads both values. The file remains on the disk after the program ends. That is the whole point of file handling.

Program 4.2 Write and read a binary student record

```

#include <iostream>
#include <fstream>
#include <cstring>
using namespace std;

struct Student {
    char name[20];
    int age;
    float percentage;
};

int main() {
    Student s;
    strcpy(s.name, "Ayesha");
    s.age = 17;
    s.percentage = 88.5f;

    ofstream outFile("student.bin", ios::binary);
    outFile.write(reinterpret_cast<char*>(&s), sizeof(s));
    outFile.close();

    Student loaded;
    ifstream inFile("student.bin", ios::binary);
    inFile.read(reinterpret_cast<char*>(&loaded), sizeof(loaded));
    inFile.close();

    cout << "Name: " << loaded.name << endl;
    cout << "Age: " << loaded.age << endl;
    cout << "Percentage: " << loaded.percentage << endl;
    return 0;
}

```

The student is a structure with a char array, an int and a float, so the object can be copied as a block of bytes. The name is placed with strcpy so the array contains a proper C string. Reading uses the same struct and the same size. If the reading program used a different layout, the bytes would be interpreted wrongly.

Exam point

State text versus binary before you write code in a long question. Then show open, the transfer, close, and a second open to read the data back. Mention EOF if the question asks how a loop knows when to stop.

Key terms

Term	Meaning to write in an answer
Text file	A file stored as characters.
Binary file	A file stored as the raw bytes of the data.
ofstream	A stream used to write a file.
ifstream	A stream used to read a file.
Append	Adding new data at the end of an existing file.
EOF	The end-of-file condition, true when no more data can be read.

Bring it together

- Files keep data after the program ends.
- Text files store characters. Binary files store memory images of objects.
- Choose in, out or app, and add binary when the bytes must be preserved.
- Open, check, read or write, and close. EOF tells an input loop to stop.

Do this in the practical

- 1 Write a name and a mark to a text file and read them back.
- 2 Write a student structure to a binary file and read it into a second variable.

Check yourself

Attempt every item before you read the key. Section A of the Computer Science paper is a set of multiple-choice questions.

1. A text file stores:

- A. Characters
- B. Only electrical addresses
- C. Flowchart symbols
- D. OSI layers

2. ios::app means:

- A. Append at the end of the file
- B. Erase ROM
- C. Sort an array
- D. Close the monitor

3. ios::out on an existing file:

- A. Opens it for writing and discards the old contents
- B. Always appends
- C. Converts it to a pointer
- D. Marks it read-only in Windows

4. EOF becomes true when:

- A. A read passes the last data
- B. The computer boots
- C. A class is declared
- D. Wi-Fi connects

5. Binary read and write transfer:

- A. A block of bytes of a given size
- B. Only one flowchart
- C. Only comments
- D. A topology name

6. Which header declares file streams?

- A. fstream
- B. iostream only, with no file support at all
- C. cmath
- D. cstring only

7. Closing a file:

- A. Flushes the buffer and releases the file
- B. Deletes the compiler
- C. Changes the address bus
- D. Is illegal in C++

8. A binary student record should be read into:

- A. An object with the same member layout
- B. A different unrelated type on purpose
- C. An OSI layer
- D. A service level agreement

Answer key

Q	Answer	Why this is the answer
1	A	Text files hold character data that an editor can show.
2	A	Append keeps existing data and adds at the end.
3	A	Output mode creates or empties the file. Append mode keeps it.
4	A	End of file means there is nothing left to read.
5	A	read and write copy a stated number of bytes.
6	A	File stream classes are declared in fstream. iostream covers cin and cout.
7	A	close completes pending output and frees the file.
8	A	The bytes match the struct that was written.

Short answers

A 3-mark answer needs a definition and two clear points, or a three-row difference. Write in sentences.

Q1. Differentiate a text file and a binary file.

Model answer

A text file stores data as characters and can be opened in an editor. A binary file stores the bytes of values as they are held in memory, using read and write. Binary storage preserves exact numeric bits and is not meant to be edited as a document.

Q2. Differentiate write mode and append mode.

Model answer

ios::out opens a file for writing and empties it if it already exists. ios::app opens a file so that new output is added after the existing contents. Use out when the file should be created fresh. Use app when older records must remain.

Q3. What are BOF and EOF?

Model answer

BOF is the beginning of the file, the position where reading normally starts. EOF is the end-of-file condition, which becomes true when a read can obtain no more data. An input loop stops when EOF is reached.

Q4. Write the statements that open a text file for output and write a name.

Model answer

ofstream outFile("student.txt"); outFile << name << endl; outFile.close(); The first statement creates or empties the file. The second writes the name. The third closes it.

Q5. Why must the struct used to read a binary record match the struct that wrote it?

Model answer

write copies the bytes of the object in memory order. read copies those same bytes back. A different member order or a different size would place the bytes into the wrong members, so the name, age or percentage would be wrong.

Detailed answers

Section C asks for a connected answer. Open with a definition, develop the points below, and close with one example.

Question 1. Explain file modes and the difference between text and binary handling. Include BOF and EOF.

- 1 Define file handling and why files outlive variables.
- 2 Contrast text and binary with one example of each transfer: << and >>, then read and write.
- 3 Explain in, out, app and binary.
- 4 Define BOF and EOF and show them in a read loop.
- 5 Mention open, test, use and close as the safe order.

Question 2. Describe a program that stores a student in a binary file and reads the student back.

- 1 Define a struct with a char name, an int age and a float percentage.
- 2 Fill the fields, using strcpy for the name.
- 3 Open an ofstream with ios::binary and write sizeof bytes.
- 4 Close, open an ifstream with ios::binary, and read into a second struct.
- 5 Print the fields to show the data survived.